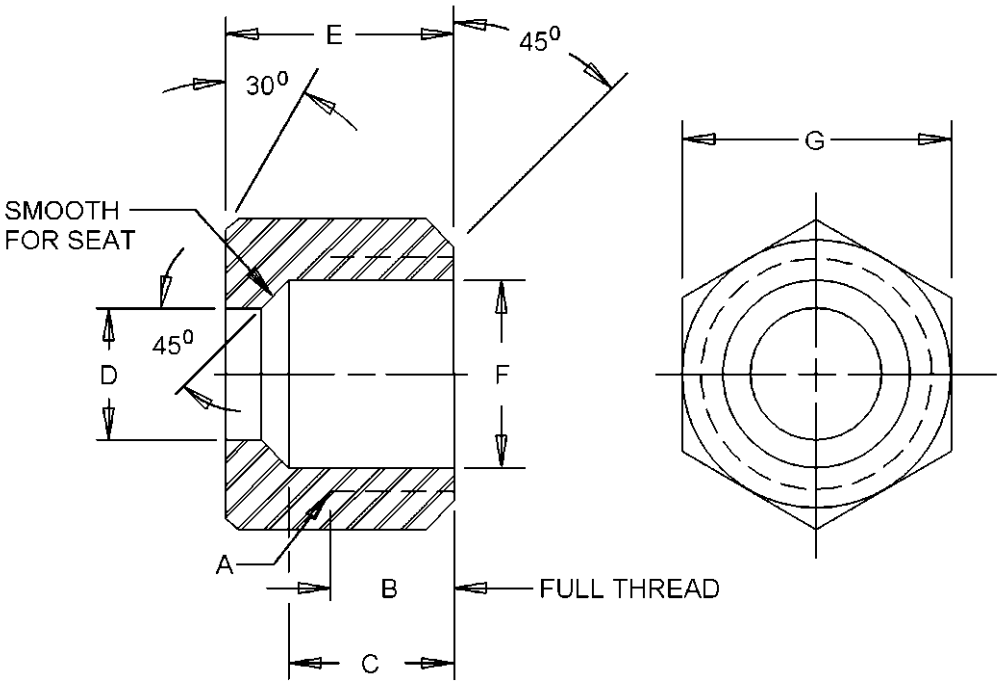
INCH-POUND
MS39210K
31 January 2018
SUPERSEDING
MS39210J
30 August 2012

DETAIL SPECIFICATION SHEET

NUTS, TUBE COUPLING – SAFETY SLEEVE COMPRESSION TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Nut, tube coupling.

AMSC N/A

FSC 4730

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Dash Number	Tube OD	A Thread (see note 4)	B Thread Length inch (mm)	C inch (mm)
-1	.1250 (3.175)	.3125-24UNF-2B	.219 (5.56)	.297 (7.54)
-2	.1875 (4.763)	.3750-24UNF-2B	.219 (5.56)	.297 (7.54)
-3	.2500 (6.350)	.5000-20UNF-2B	.281 (7.14)	.359 (3.13)
-4	.3125 (7.938)	.5625-8UNF-2B	.281 (7.14)	.390 (3.92)
-5	.3750 (9.525)	.6250-18UNF-2B	.313 (7.94)	.422 (10.72)
-6	.5000 (12.700)	.7500-16UNF-2B	.328 (3.33)	.453 (11.51)
-7	.6250 (15.875)	.9375-16UN-2B	.375 (9.53)	.469 (11.91)
-8	.7500 (19.050)	1.0625 16UN-2B	.406 (10.32)	.500 (12.70)
-9	.8750 (22.225)	1.1875-16UN-2B	.406 (10.32)	.500 (12.70)
-10	1.0000 (25.400)	1.312516UN-2B	.406 (10.32)	.500 (12.70)
-11	1.2500 (31.750)	1.6250-16UN-2B	.406 (10.32)	.578 (14.68)
-12	1.5000 (38.100)	1.8750-16UN-2B	.500 (12.70)	.672 (17.07)

Dash Number	D ± .005 (0.13) inch (mm)	E inch (mm)	F ± .005 (0.13) inch (mm)	G inch (mm)
-1	.130 (3.30) .134(3.40)	.438 (11.11)	.275(6.99) .270(6.86)	.438 (11.11)
-2	.190(4.83) .194(4.93)	.438 (11.11)	.338(8.59) .333(8.46)	.500 (12.70)
-3	.255(6.48) .259(6.58)	.563 (14.29)	.454(11.53) .449(11.40)	.625 (15.88)
-4	.318(8.08) .322(8.18)	.578 (14.68)	.512(13.00) .505(12.83)	.688 (17.46)
-5	.380(9.65) .384(9.75)	.625 (15.88)	.575(14.61) .568(14.43)	.750 (19.05)
-6	.506(12.85) .510(12.95)	.688 (17.46)	.693(17.60) .686(17.42)	.875 (22.23)
-7	.633(16.08) .637(16.18)	.750 (19.05)	.881(22.38) .874(22.20)	1.125 (28.58)
-8	.758(19.25) .762(19.35)	.813 (20.64)	1.005(25.53) .999(25.37)	1.250 (31.75)
-9	.883(22.43) .887(22.53)	.813 (20.64)	1.131(28.73) 1.124(28.55)	1.375 (34.93)
-10	1.008(25.60) 1.012(25.70)	.813 (20.64)	1.256(31.90) 1.249(31.72)	1.500 (38.10)
-11	1.206(30.63) 1.264(32.11)	.844 (31.43)	1.568(39.83) 1.561(39.65)	2.000 (50.80)
-12	1.510(38.35) 1.514(38.46)	.969 (34.61)	1.818(46.18) 1.810(45.97)	2.250 (57.15)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, the tolerance is ±.016 (0.41 mm), the tolerance for degrees is ± 2°.
4. Threads shall be in accordance with FED-STD-H28.
5. This illustration is for identification and is not intended to restrict designs or shapes not dimensioned.

FIGURE 1. Nut, tube coupling - Continued.

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REQUIREMENTS:

Fittings shall be as specified on figure 1 and table I.

Materials shall be in accordance with MIL-DTL-18866 and table I. Components of the assemblies shall be of the same material and finish.

Working pressure is the established recommended pressure based on a safety factor of 4 to 1, see MIL-DTL-18866.

Finish. Finishes shall be as specified in table I. All plating's shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

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Table I. Material and finish designators.

Material finish designator	Material	Finish
C	Steel bar, type 1117, 1118 or 1137 in accordance with ASTM A29/A29M	Carbon steel: Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick. 1/
CZ	Steel bar, type 1117, 1118 or 1137 in accordance with ASTM A29/A29M	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 2/
F	Steel	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick. 1/
FZ	Steel	Zinc plating in accordance with ASTM B633; type III or V, Fe/Zn 8. 2/
H	Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 3/
J	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 3/ 4/
M	Nickel-copper alloy UNS N04400	No additional finish.
N	High-chromium nickel alloy UNS N06690	No additional finish.
P	Steel	Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 2/
S	Corrosion resistant steel	Corrosion resistant steel: Passivate in accordance with SAE AMS2700, type 6 or 7.
T	Titanium	Anodize in accordance with SAE-AMS2488 type 2. 5/
TF	Titanium	Fluoride phosphate in accordance with SAE-AMS2486. 5/
Z	Steel	Zinc plating in accordance with ASTM B695, type II, class 8 2/
ZC	Steel	Zinc may be any zinc plating from PIN codes CZ, J, and Z with a colored chromate coating. 2/

1/ Embrittlement test need not be run. Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA owned shipboard systems.
2/ Not for use in aircraft. Requires approval from the Program Office for all applications.
3/ Hexavalent chromium free. Finish shall be RoHS compliant.
4/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293. Titanium shall not be used in oxygen or potable water systems.

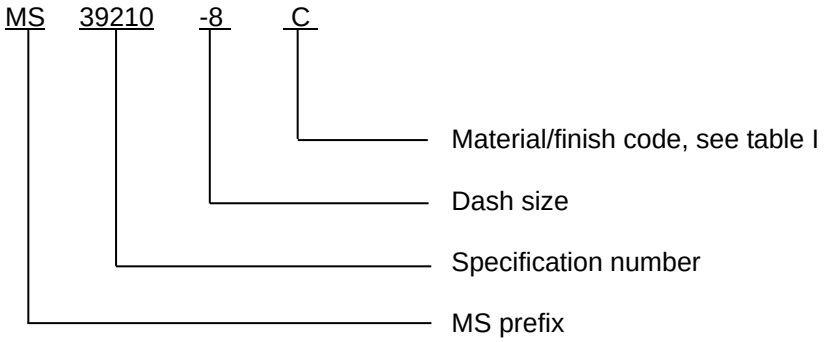
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Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using one of the brush plating processes below as appropriate to the primary finish. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS-2451/5.

b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS-2451/9.

Part or Identifying Number (PIN) example:



MS39210-8C indicates a nut for a .750 (19.05 mm) inch tube, steel, cadmium plated.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides an alternate material, corrosion resistant steel, and zinc finish via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Cadmium is prohibited from use in NAVSEA owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2.

Class I and II ozone depleting substances (ODS) shall not be used in MS39210 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-18866, this document references the following:

FED-STD-H28	SAE-AMS-STD-595/36099	SAE-AMS-STD-595/36280
MIL-DTL-16232	SAE-AMS-STD-595/36118	SAE-AMS-STD-595/36293
ASTM A29/A29M	SAE-AMS-STD-595/36134	SAE-AMS2417
ASTM B117	SAE-AMS-STD-595/36152	SAE-AMS2451/5
ASTM B633	SAE-AMS-STD-595/36170	SAE-AMS2451/9
ASTM B695	SAE-AMS-STD-595/36173	SAE-AMS2486
ASTM F1136/F1136M	SAE-AMS-STD-595/36176	SAE-AMS2488
SAE-AMS-QQ-P-416	SAE-AMS-STD-595/36231	SAE-AMS2700
SAE-AMS-STD-595/36076	SAE-AMS-STD-595/36251	T9070-AL-DPC-020/077-2
SAE-AMS-STD-595/36081	SAE-AMS-STD-595/36270	

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - YD
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2018-017)

Review activities:
Army - AR
Navy - MC, SA
Air Force - 71

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